

- (1) At the cathode, Electrical energy gained is converted into Kinetic Energy. ✓ AN

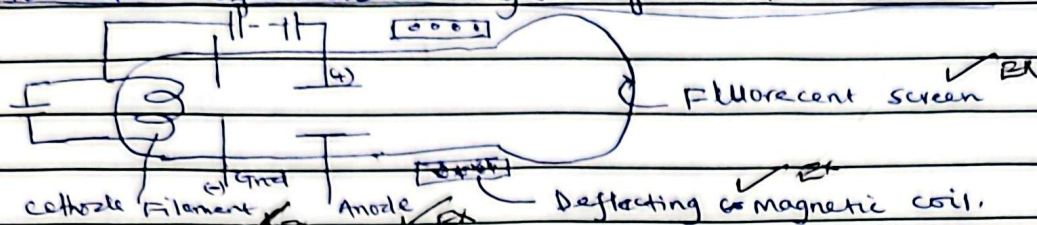
$$eV = \frac{1}{2}mv^2 \quad \checkmark \text{ AN}$$

$$(1.6 \times 10^{-19} \times 8.0 \times 10^3) = \frac{1}{2} \times (9.11 \times 10^{-31}) \cdot v^2 \quad \checkmark \text{ AN}$$

$$v = 5.3 \times 10^7 \text{ m s}^{-1} \quad \checkmark \text{ AN}$$

The screen gives normal brightness if electrons strike the screen with speed of $5.3 \times 10^7 \text{ m s}^{-1}$; this condition is satisfied, thus the screen gives normal brightness. ✓ EV

Production of visible images by a CRT Television.



- The cathode filament is heated and releases electrons by thermionic emission. ✓ EX
- The positive anode accelerates the electrons towards the screen. ✓ EX
- The control grid controls the amount of electrons accelerated to the screen, and control brightness. ✓ EX
- Magnetic coils located on the outside of the tube neck generate magnetic fields that bend the electron beam to the exact position on the screen. ✓ EX
- The inside of the front screen is coated with phosphor, material that glows when hit by electrons. ✓ EX
- The beam sweeps across the screen, horizontally and then vertically, at high frequency, creating image. ✓ EX

Brightness control:

When light falls on photo sensitive metal plate, electrons are emitted from the metal surface by photoelectric emission. The emitted electrons produce a current that is sent to the control grid. ✓ EX

In bright light, more electrons are emitted, photocurrent increases. The grid voltage increases; More electrons are directed to the screen and brightness increases.

In weak light, few photoelectrons are emitted, photocurrent reduces, grid voltage reduces, Less electrons reach the screen and it becomes dimmer.

Brightness remaining the same in sharp orange-red light.

Photon Energy $E = hf$ ✓ AN

$$E = 6.63 \times 10^{-34} \times 2.9 \times 10^{14}$$
 ✓ AN

$$E = 1.9227 \times 10^{-19} \text{ J}$$

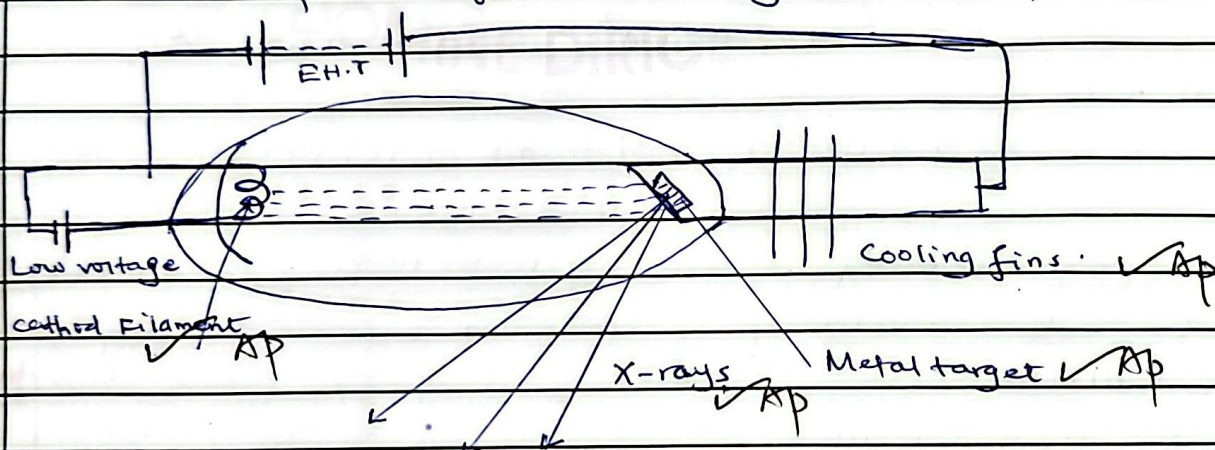
$$\therefore E = \frac{1.9227 \times 10^{-19}}{1.6 \times 10^{-19}} = 1.2 \text{ eV.} \quad \checkmark \text{ AN}$$

Work function $\phi = 1.56 \text{ eV.}$

Since $1.2 \text{ eV} < \phi$, electrons are not emitted, no photocurrent flows to the an. grid and the Brightness on the screen remains the same. ✓ EV

(b) Any other device that applies moving electrons:

Production of X-rays in an X-ray tube. ✓ AP

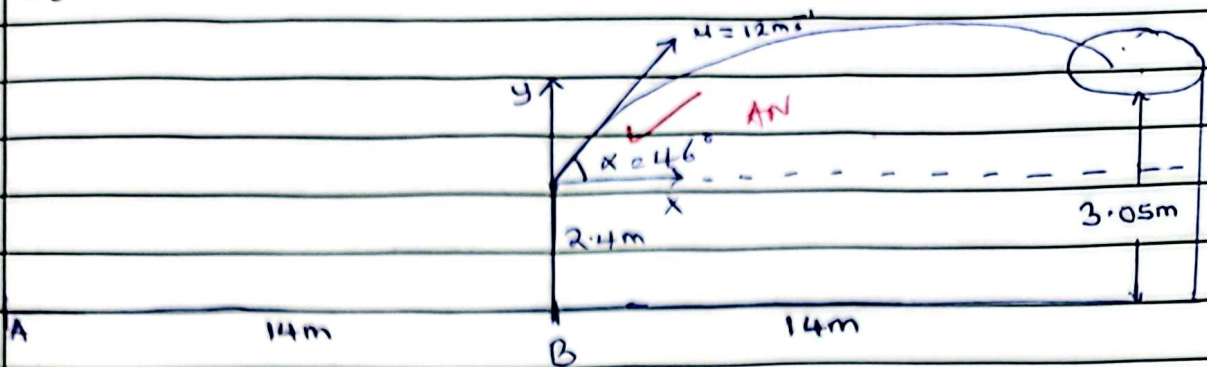


Low voltage heats filament. electrons are produced by thermionic emission. High voltage accelerates the electrons to the anode.

The fast moving electrons are stopped by the metal target. about 1% of the electron energy is converted to X-rays and the remaining energy is converted into heat. ✓ AP

(2)

(a)



Assuming that Neglecting the horizontal velocity of the player at the time of projecting the ball,

$$(u \cos \alpha) t = 14 \quad \checkmark \quad \text{AN}$$

$$t = \left(\frac{14}{12 \cos 46^\circ} \right) = \frac{14}{8.3359} = 1.68 \text{ s} \quad \checkmark \quad \text{AN}$$

Time taken before projection: $(5.2 - 1.68) = 3.52 \text{ s} \quad \checkmark \quad \text{AN}$

The ball was projected 3.52 s after starting to sprint

Since $3.52 \text{ s} < 4.0 \text{ s}$ ~~EV~~ The ball was projected before the buzzer. Thus the score counted (Valid). ~~EV~~

Motion of the player:

Uniform acceleration: $u = 0$, $S = 14 \text{ s}$, $t = 3.52 \text{ s}$.

$$S = ut + \frac{1}{2} at^2 \quad \checkmark \quad \text{AN}$$

$$14 = 0 + \frac{1}{2} a (3.52)^2, \quad \checkmark \quad \text{AN} \quad a = 2.26 \text{ m/s}^2 \quad \checkmark \quad \text{AN}$$

Final Velocity:-

$$V = u + at, \quad \checkmark \quad \text{AN} \quad V = 0 + (2.26) \times (3.52) \quad \checkmark \quad \text{AN}$$

$$V = 7.9552 \text{ m/s} \approx 8.0 \text{ m/s} \quad \checkmark \quad \text{AN}$$

The player starts from rest accelerates uniformly at 2.26 m/s^2 and reaches a velocity of 8 m/s just before releasing the ball. ~~AN~~

Motion of the ball:-

The ball moves in a parabolic path. ~~EXB~~

The ball ~~is~~ moves with a constant horizontal velocity, and a uniform vertical ~~EX~~ acceleration (acceleration due to gravity)

A constant vertical force (mg) constantly pulls the ball ~~EX~~ down wards. This makes the ball move along a parabolic path.

Name.....Class/Stream.....

Paper No.Subject

Signature Date: Time:

Do not
write
in this
margin

b) David's closing remarks:-

Jump shot:

The player holds a stationary position, the player's horizontal velocity is zero. The ball is then only projected and moves with intended velocity. If the player has a horizontal velocity, the actual velocity of the ball would be the sum of the player's velocity and velocity of projection of the ball. This would increase horizontal velocity of the ball, and accuracy would reduce.

A Follow through helps to provide extended time during throwing. The ball gets enough required momentum.

In a controlled way: From, $\text{Impulse} = \text{Force} \times \text{time}$.

The ball attains increased velocity with minimum force and the velocity is in a particular intended direction.

Snap shot creates a back spin.

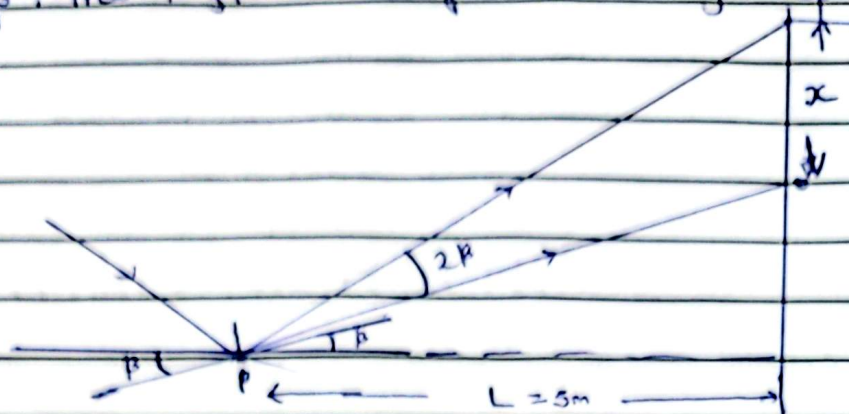


As the ball spins in air, there is increased velocity of air above the ball, the pressure there reduces.

The reduced air velocity below the ball then increases the pressure. The ball experiences a small uplift due to the pressure difference. The ball keeps in air longer and moves at a controlled speed. This improves the accuracy of to the target.

3)

(c) When the reflecting water surface is tilted through a small angle β , the reflected ray turns through 2β . ✓ AN



For small angles in radians, ✓ AN $\tan 2\beta \approx 2\beta = \frac{x}{L}$

$$\beta = \frac{x}{2L} = \frac{0.8}{2 \times 5} = 0.08 \text{ radians} \quad \checkmark \text{ AN}$$

The Lake wave is $y = A \sin(2\pi ft - kx)$

$$\left| \frac{dy}{dx} \right|_{\text{max}} = |Ak| = \beta \quad \checkmark \text{ AN}$$

$$\text{Wave Number: } k = \frac{2\pi}{\lambda} = \left(\frac{2\pi}{20} \right) = \left(\frac{\pi}{10} \right) \text{ rad m}^{-1} \quad \checkmark \text{ AN}$$

$$\text{Now:- Amplitude: } A = \frac{\beta}{k} = \frac{0.08}{\frac{\pi}{10}} = (0.255 \text{ m}) \quad \checkmark \text{ AN}$$

The device can generate power if $A > 0.25 \text{ m}$. ✓ EV

Since $A = 0.255 \text{ m} > 0.25 \text{ m}$, ✓ EV The project is supported by the lake waves. ✓ EV

The Wave Equation:

$$T = 5.0 \text{ s}, \quad f = \frac{1}{T} = \frac{1}{5.0} = 0.20 \text{ Hz} \quad \checkmark \text{ AN}$$

$$\text{Using } y = A \sin(2\pi ft - kx)$$

$$y = 0.255 \sin(2\pi(0.2)t - \frac{\pi x}{10}) \quad \checkmark \text{ AN}$$

$$\therefore y = 0.255 \sin(0.4\pi t - \frac{\pi x}{10}) \quad \checkmark \text{ AN}$$

Name.....Class/Stream.....

Paper No.Subject

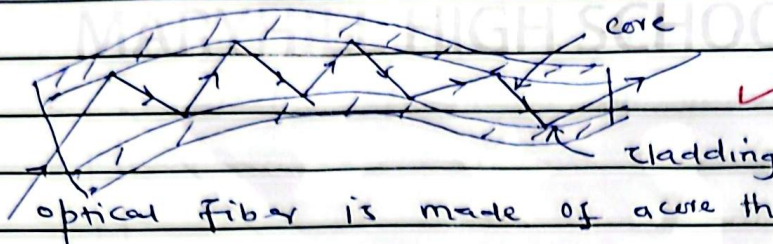
SignatureDate:Time:

The convex lens use:

(b) The reflected beam from the wave may spread. The convex lens then collects the light and converges them. ✓ EX

The beam is then focused into the small opening of the optical fiber. Almost all the reflected light is focused in the optical fiber, this improves signal strength. ✓ EX

Optical fiber signal transmission:-



Optical fiber is made of a core that has more refractive index than the cladding.

Light is directed at the boundary between the core and the cladding at an angle greater than the critical angle. ✓ AP

The light is totally internally reflected, the same happens inside the tube at points of incidence. Multiple total internal reflections helps to transmit light through the optical fiber without loss of intensity. ✓ AP

This method is accurate because it reduces human errors, improves sensitivity. ✓

4)

- a) A concave mirror reflects parallel rays onto the focal point. ✓ ~~EX~~
- The concave mirror is made to face directly towards the sun. ✓ ~~EX~~
- The blackened metal container is moved along the principal axis of the mirror. ✓ ~~EX~~
- The position is adjusted until the reflected light forms a smallest and brightest spot on the container. ✓ ~~EX~~
- This position corresponds to the focal point.
- The blackened container then absorbs the radiation reflected from the mirror.

Determining if the system is efficient for use:

Solar power incident on the mirror: $P = IA$ ✓ ~~AN~~ $A = \pi r^2$

Intensity $I = 1.4 \times 10^3 \text{ W m}^{-2}$ ✓ ~~AN~~

$$P = (1.4 \times 10^3) \times \pi (1.2)^2 = 6328 \text{ W} \cdot \checkmark \text{ AN}$$

65% Reaches the blackened container.

$$P = \left(\frac{65}{100}\right) \times 6328 = 4113 \text{ W} \cdot \checkmark \text{ AN}$$

$\therefore$ Power heating water = $4.113 \times 10^3 \text{ W}$.

Heat needed to convert water ice at -5°C to water at 80°C

$Q = \text{Heat to melt ice from } -5^\circ\text{C to } 0^\circ\text{C} + \text{Heat to melt ice} + \text{Heat } 0^\circ\text{C to } 80^\circ\text{C}$

$$Q = MC_i(0 - (-5)) + ML_f + MC_w(80 - 0)$$

$$Q = 5 \times 2.1 \times 10^3 \times 5 + 5 \times 3.3 \times 10^5 + 5 \times 4.2 \times 10^3 \times 80 \cdot \checkmark \text{ AN}$$

$$Q = 52500 + 1.65 \times 10^6 + 1.68 \times 10^6$$

$$Q = 3.3825 \times 10^6 \text{ J} \cdot \checkmark \text{ AN}$$

$$\text{Thus, } Q = P \cdot t, \quad t = \frac{3.3825 \times 10^6}{4.113 \times 10^3} \cdot \checkmark \text{ AN}$$

$$t = 822.8$$

$$\text{Convert to minutes: } t = \frac{822.8}{60} = 13.7 \text{ min} \cdot \checkmark \text{ AN}$$

Name..... Class/Stream.....

Paper No. Subject

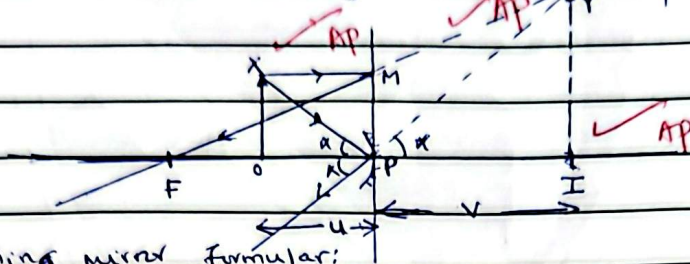
Signature Date: Time:

Do not
write
in this
margin

Since single session period is 40min, Needed time of 13.7min is less than 40min, the system can be efficient for use.

(b) A dentist needs a large clear view of a small tooth surface. When the tooth is placed between the focus and pole of the concave mirror, the image becomes enlarged, the image remains upright and the image is Virtual. The dentist can observe details more clearly. The ray diagram below shows the action of the mirror.

The image formed is virtual magnified and upright.



Establishing mirror formula:

Triangles OXP and PYI are similar.

$$\frac{YI}{OX} = \frac{-v}{u} \quad (i)$$

Triangles FMP and FYI are similar:

$$\frac{YI}{MP} = \frac{(-v + f)}{f}$$

Since $OX = MP$:

$$\frac{-v}{u} = \frac{-v + f}{f}$$

$$-vf = -uv + uf$$

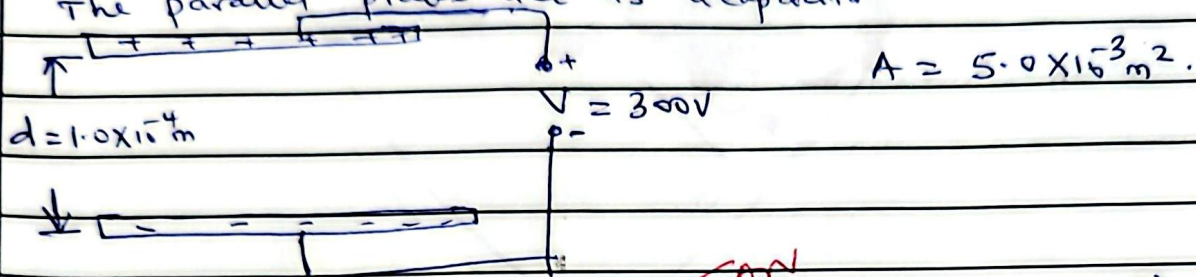
$-uv = vf + uf$, Dividing by uvf .

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

(5)

- (a) - An ink jet printer ejects tiny droplets of ink from an nozzle.
- The droplets are charged negatively by the charging system, then pass between two oppositely charged parallel metal plates.
 - Electric field between the plates exerts a force on each charged droplet.
 - This force deflects the droplets so that they land at the required position on the moving paper.
 - correct printing depends on enough charge on the droplets, and strong enough electric field between the plates.

To check for sufficient charge on the plates;
The parallel plates act as a capacitor.



$$\text{Capacitance } C = \frac{\epsilon_0 A}{d} = \frac{8.85 \times 10^{-12} \times 5.0 \times 10^{-3}}{1.0 \times 10^{-4}} \text{ F}$$

$$C = 4.425 \times 10^{-10} \text{ F}$$

Charge on the plates:

$$Q = CV, \quad Q = (4.425 \times 10^{-10}) \times 300 = 1.33 \times 10^{-7} \text{ C}$$

For efficient printing, charge on the plates is $1.4 \times 10^{-7} \text{ C}$.
Charge on the plates $1.33 \times 10^{-7} \text{ C} < 1.4 \times 10^{-7} \text{ C}$ the charge is not sufficient.

To check for force on the droplets:

$$\text{Force: } F = q \cdot E$$

$$\text{Electric field intensity } E = \frac{V}{d} = \frac{300}{1.0 \times 10^{-4}} = 3.0 \times 10^6 \text{ N/C}$$

$$\text{Force on one droplet: } F = q \cdot E = (4.0 \times 10^{-13} \times 3.0 \times 10^6) \text{ N}$$

Paper No. Subject

Signature Date: Time:

$$F = 1.2 \times 10^6 \text{ N},$$

For Efficient printing, $F > 1.2 \times 10^6 \text{ N}$. Since $F = 1.2 \times 10^6 \text{ N}$,
the force is not sufficient for proper printing. ✓

Advise to the technician,

The deflecting system should be checked, such that
the P.d across the plates is increased;

(b) Length of Nichrome wire needed to improvise the
resistor:

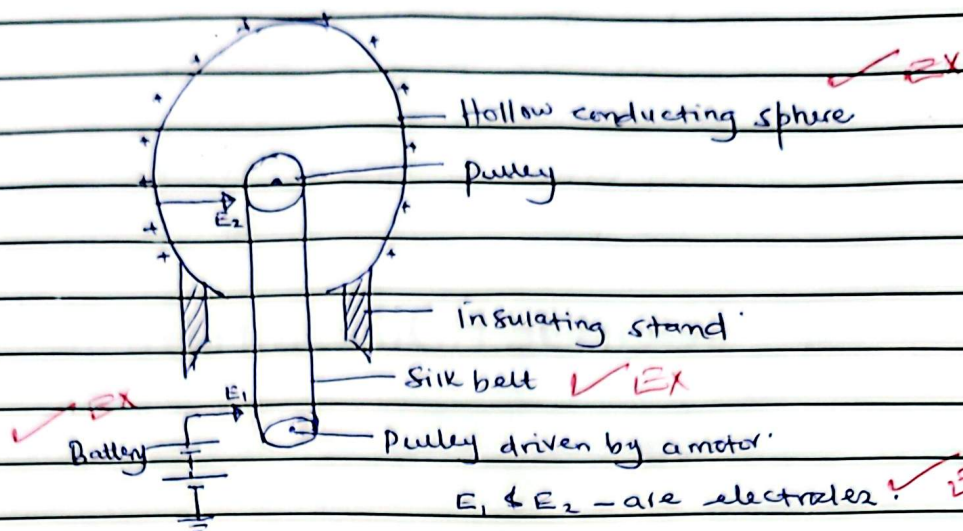
$$V = 120 \text{ V}, \quad I = 0.15 \text{ A} \quad R = \frac{V}{I}, \quad R = \frac{120}{0.15}, \quad R = 800 \Omega \quad \text{✓ AN}$$

$$R = \frac{\rho l}{A}, \quad l = \frac{R \cdot A}{\rho} = \frac{800 \pi (5 \times 10^{-5})^2}{1.0 \times 10^{-6}} \quad \text{✓ AN}$$

$$l = 6.28 \text{ m} \quad \text{✓ AN}$$

The required wire length is 6.28 m.

(6)(a) Working of a Van de Graaff generator.

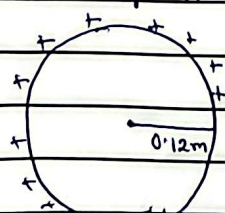


- Electrode E_1 is made positive relative to the earth, at E_1 there is high charge density and high electric field intensity, Air around E_1 gets ionised.
- Positive ions are repelled to the belt which drives them up to the sphere. At the sphere, Negative charge is induced at E_2 and positive charge induced on the surface of the sphere.
- Point action takes place at E_2 , Negative ions are repelled to the belt and discharge it before it moves over the pulley.
- The process continues until the charge on the sphere can ionise the air around it.

Decision to accept the generator or not:-

Energy supplied by the motor: $E = V I t = 12 \times 0.5 \times 60$
 $E = 360 \text{ J}$

Energy stored by the sphere:



$R = 2.4 \times 10^{-6} \text{ C}$

Capacitance: $C = 4\pi\epsilon_0 R$

$C = 4\pi \times 8.85 \times 10^{-12} \times 0.12$

$C = 1.33 \times 10^{-11} \text{ F}$

$$\text{Electrostatic Energy } U = \frac{q^2}{2\epsilon} = \frac{(2.4 \times 10^{-6})^2}{2 \times 1.33 \times 10^{-11}} \quad \checkmark \text{ AN}$$

$$U = 0.2165 \quad \checkmark \text{ AN}$$

Efficiency of conversion of energy;

$$\eta = \frac{0.216}{360} \times 100\%$$

$$\eta = 0.06\% \quad \checkmark \text{ AN}$$

Required efficiency of Energy conversion; is $0.05\% < 0.06\%$. ✓ EV

For the Van de Graaff Generator to be accepted, the sphere must be smooth, spherical and clean. Thus charge must be uniformly distributed over the surface. To check for this, ✓ EV

- A proof plane is placed on the surface of the dome that is charged, and then transferred into a metal can.

placed on the cap of a neutral gold leaf electroscope. ✓ EX

- Divergence on the gold leaf electroscope is noted. ✓ EX

- The experiment is repeated with the proof plane placed at different parts of the dome, and corresponding divergencies noted. ✓ EX

- If the divergencies are equal, the charge is then uniformly distributed. ✓ EX

Since Efficiency of energy conversion is greater than 0.05% , if the charge is uniformly distributed over the dome, the Generator can be accepted. ✓ EX

(b)

As the Van de Graaff is kept running, charge on the dome kept on increasing until a value when the insulation of air breaks down. ✓ AN

$$\text{Since } E = \frac{Q}{4\pi\epsilon_0 r^2}$$

$$Q = (3.0 \times 10^6) \cdot 4\pi\epsilon_0 r^2$$

$$Q = 3.0 \times 10^6 \cdot 4\pi \times 8.85 \times 10^{-12} (0.12)^2$$

$$Q = 4.8 \times 10^{-6} \text{ C} \quad \checkmark \text{ AN}$$

Do not
write
in this
margin

Name.....Class/Stream.....

Paper No.Subject

Signature Date: Time:

Do not
write
in this
margin

When the charge increases to $4.8 \times 10^{-6} \text{ C}$ the electric field intensity is high enough to break down insulation of the air around the sphere, thus causing sparks around the dome.